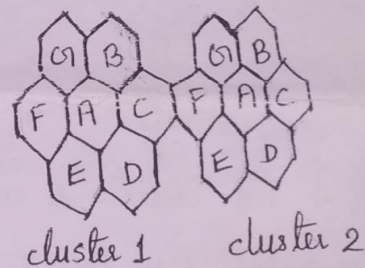
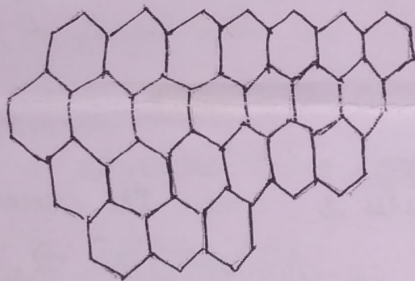


Mobile Communication [Cellular Communication]

Cell Concept

- In mobile communication, the entire geographical area is divided into a number of cells.
- The cells are hexagonal in shape ideally. The cells are organised into clusters with most cellular systems using seven cells in the cluster.
- Each cluster is allocated a fixed band of frequencies. These frequencies are divided among cells in a cluster. Adjacent cells cannot use same frequency.
- The cell structure is given below:

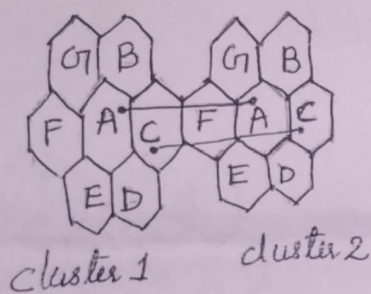


- Each cell contains an antenna and is controlled by a small office within the cell called cell office [Base Trans receiver Station, BTS]
- All subscribers inside a particular cell are monitored and tracked continuously by the BTS and data is passed on to the central office [Mobile Switching Centre, MSC].
- MSC records all data regarding all calls, billing of calls etc.
- The area of each cell will depend on the number of subscribers,

density of the buildings, geographical area etc. High population density areas require geographically smaller cells to meet the traffic demands.

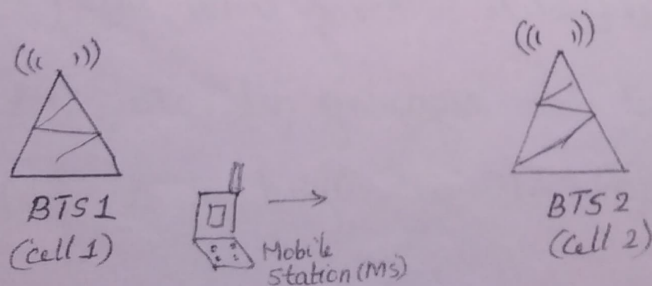
Frequency Reuse

- It is the process in which the same set of frequencies can be allocated to more than one cell provided the cells are separated by sufficient distance.
- In cellular communication system, frequency reuse allows a service provider to increase the number of subscribers.
- Consider two clusters of seven cells each:



- cell A in cluster 1 and cluster 2 uses the same set of frequencies. Thus they are called co-channel cells. The same set of frequencies are reused in co-channel cells but there must be sufficient distance between them.
- cells A & B use different set of frequencies and are called adjacent cells.

Hand over / Hand off



- When a mobile handset ^(mobile station, MS) moves from cell 1 to cell 2, the signal strength obtained for that mobile from cell 1 begins to decrease and the signal strength from cell 2 goes on increasing.
- When the signal strength from cell 2 increases above a threshold value and the signal strength from cell 1 decreases below a threshold value, the mobile handset initiates hand off.
- This time the cell 2 allocates a channel for the mobile set to continue the call and the channel for the call from cell 1 gets disconnected. The mobile handset will continue the call through the newly allocated channel. This process is called hand off or hand over.

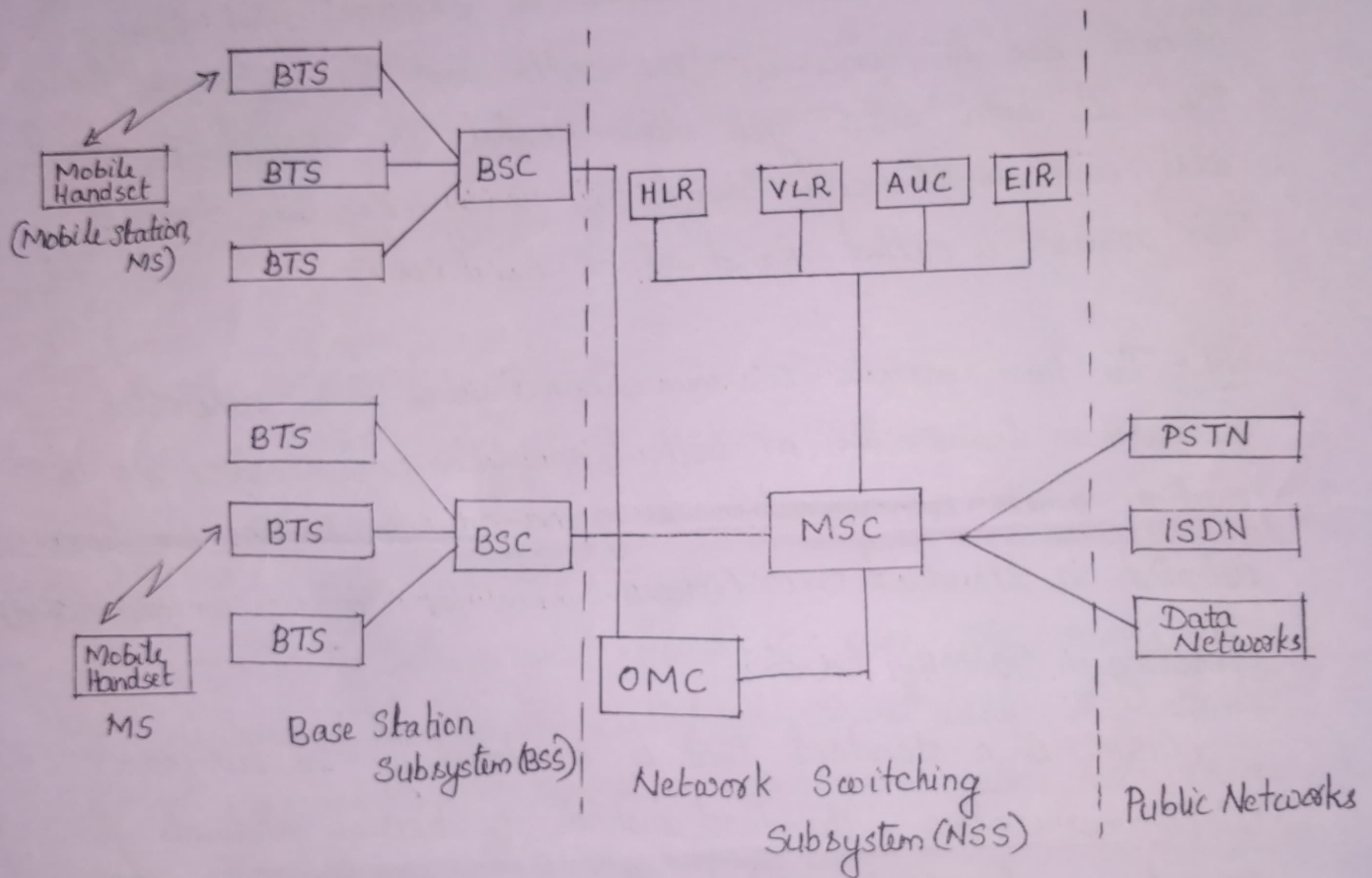
Note: The term mobile communication implies that either the receiver or transmitter or both transmitter and receiver are in motion while communication is going on. In India, we have adopted a standard GSM (Global System for Mobile communication) operating in 900MHz band.

- GSM is a standard that is developed by the European telecommunication standard institute to describe protocols for second generation (2G) digital cellular networks used by mobile phones.
- Frequency band of 890MHz - 915MHz is the uplink frequency and 935MHz - 960MHz is the downlink frequency used in GSM.

GSM Architecture

A GSM network consists of mainly three parts:

- (a) Mobile station (MS)
- (b) Base Station Subsystem (BSS)
- (c) Network Switching Subsystem (NSS)



BTS - Base Trans receiver Station

BSC - Base Station Controller

HLR - Home Location Register

VLR - Visitor Location Register

AUC - Authentication Centre

EIR - Equipment Identity Register

MSC - Mobile Switching Centre

OMC - Operation & Maintenance Centre

PSTN - Public Switched Telephone Network

ISDN - Integrated Services Digital Network

Mobile Station

- It consists of mobile equipment such as cellphone and a smart card called Subscriber Identity Module or SIM, which contains entire customer related information.

- Each handset is having a unique identity number known as International Mobile Equipment Identity (IMEI) number which is installed in the phone while manufacturing and that cannot be changed. An IMEI is marked as invalid if it has been reported stolen.

- SIM card is a memory card where informations like SIM number, International Mobile Subscriber Identity number (IMSI), a secret key for its authentication etc. are there.

- The SIM card is protected against the unauthorised use by a 4-digit password known as Personal Identification Number (PIN).

(b) Base Station Subsystem (BSS)

The BSS mainly consists of 2 parts, Base Trans receiver System (BTS) and Base Station Controller (BSC).

(i) Base Trans receiver Station (BTS)

- This is the GSM part that communicates with the mobile.

- The BTS is usually located at the center of the cell. The antenna is also included as a part of BTS.

(ii) Base Station Controller (BSC)

- A group of BTS's are controlled by a BSC.

- It provides control for the entire BSS.

- It manages radio channel set up and hand-overs.

- It provides the connection between mobile station (MS) and MSC.

(c) Network Switching Subsystem (NSS)

NSS controls the entire network. NSS contains the MSC and various databases which are used to control the network.

(i) Mobile Switching Centre (MSC)

- It is the central component of NSS and provide all functionalities needed to handle a mobile subscriber such as
 - manages mobile location
 - switches call
 - manages security function
 - control handover between BSC's etc.
- It also provides the connection to fixed network such as PSTN. (landline)

(ii) Home Location Register (HLR)

- It contains information of each subscriber registered in the corresponding GSM network, along with the current location of the mobile.

(iii) Visitor Location Register (VLR)

- It is only a temporary storage where the details are held as long as the subscriber is within the service area.
- It contains information of every visiting user in the area.

(iv) Authentication Centre (AUC)

- It generates and stores authentication parameters for subscriber identification.

(v) Equipment Identity Register (EIR)

- Valid IMEI is stored in this register.

(vi) Operations and Maintenance Centre (OMC)

- It controls and monitors other network units
- Two OMC's are used for BSS monitoring and MSC.

Establishing a call

(a) Mobile to a fixed network.

- * Destination number is dialed and call button is pressed.
- * Then the information is passed to a MSC through BSC.
- * MSC verifies the HLR of the calling number, whether it is authorized to make a call. If it is authorized, the called number will be analyzed.
- * If it is a landline number, MSC locates a path to landline number through PSTN network.
- * If the dialed phone is on hook, mobile unit receives a ring back.

(b) Mobile to Mobile.

- * Destination number is dialed and call button is pressed.
- * Then the information is passed to MSC through BSC.
- * MSC verifies the HLR of the calling number, whether it is authorized to make a call. If it is authorized, the called number will be analyzed.
- * When the number dialed is mobile, MSC finds the HLR of the called mobile number.
- * It receives the VLR ID from the HLR data and the call is routed to the corresponding MSC where the handset is present now.
- * If the destination mobile is idle, call will be connected and ring back is sent to the calling number.

(c) Fixed Unit to Mobile unit

- * The destination number is dialed and the call is routed to MSC through PSTN network.
- * MSC finds the HLR of the called mobile number.
- * It receives the VLR ID from the HLR data and the call is routed to the corresponding MSC where the handset is present now.